

SOC Analyst Lab Project

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Introduction

This project demonstrates practical SOC Level 1 skills through the simulation and investigation of multiple attack techniques using a SIEM environment. The objective is to analyze system behavior, identify suspicious activities, and correlate events based on the MITRE ATT&CK framework

Lab Environment

- The lab environment consists of:
- Windows virtual machine (target)
- Ubuntu machine running Elastic SIEM
- PowerShell for command execution
- Atomic Red Team (used in some simulations)

Logs were collected and analyzed using the SIEM to identify potential malicious activity.

Although executed in a controlled lab environment, the techniques were correlated to simulate a realistic multi-stage attack scenario.

1. Persistence Phase

The Persistence phase involves techniques used by an attacker to maintain access to a system after initial compromise.

This simulation demonstrates how a system configuration change can allow repeated access without re-exploitation.

Observed Event (SIEM)

Timestamp: Apr 23, 2026 @ 22:02:15.830

User: ASD\oolai

Process: reg.exe

SHA256: c0e25b1f9b22de445298c1e96ddfcead265ca030fa6626f61a4a4786cc4a3b7d

Command Line: "C:\Windows\system32\reg.exe" add

HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v persistence /t REG_SZ /d
"cmd.exe /c whoami" /f

Event Analysis

The execution of reg.exe indicates a modification in the Windows registry. The targeted registry path (CurrentVersion\Run) is commonly abused by attackers to establish persistence, as it allows automatic execution of programs when the user logs in.

The parent process was identified as powershell.exe, suggesting manual or scripted execution. The use of the whoami command within the registry value is unusual and may indicate reconnaissance combined with persistence.

This behavior is consistent with persistence techniques used to maintain long-term access.

MITRE ATT&CK Mapping Tactic:

Persistence (TA0003)

Technique:

T1547.001 – Registry Run Keys / Startup Folder

Detection Rules

Basic Rule: process.name: "reg.exe"

Improved Rule: process.name: "reg.exe" AND process.command_line:
"CurrentVersion\Run"

Advanced Rule: process.name: "reg.exe" AND process.command_line: "Run" AND
process.command_line: ("cmd.exe" OR "powershell.exe")

Conclusion

This activity was classified as a True Positive, demonstrating a clear persistence mechanism through registry modification.

2. Credential Access Phase

The Credential Access phase involves techniques used to retrieve stored credentials from a system. In this simulation, the Windows Credential Manager was accessed using rundll32.exe with keymgr.dll. Command Executed rundll32.exe keymgr.dll,KRShowKeyMgr

Observed Event (SIEM)

Timestamp: Apr 23, 2026 @ 23:34:12.508

Host: asd User: oolai

Process: rundll32.exe

Path:C:\Windows\System32\rundll32.exe

Command Line: "C:\Windows\system32\rundll32.exe" keymgr.dll,KRShowKeyMgr

SHA256: 076592ca1957f8f357cc201f0015072c612f5770ad7de85f87f254253c754dd7

Event Analysis

The execution of rundll32.exe with keymgr.dll indicates access to the Windows Credential Manager.

This is uncommon for regular users and may expose sensitive stored credentials.

The binary was verified as legitimate, confirming a Living off the Land (LotL) technique.

This event was correlated with persistence activity, suggesting a multi-stage attack.

MITRE ATT&CK Mapping Tactic:

Credential Access (TA0006)

Technique: T1555 – Credentials from Password Stores Detection

Detection Rules

Rules Basic Rule:

process.name: "rundll32.exe"

Improved Rule:

process.name: "rundll32.exe" AND

process.command_line: "keymgr"

Advanced Rule:

process.name: "rundll32.exe" AND

process.command_line: "keymgr.dll" AND

process.command_line: "KRShowKeyMgr"

Conclusion

This activity was classified as a True Positive and demonstrates credential access using trusted system binaries.

3. Discovery Phase

The Discovery phase involves gathering information about the system, user, and running processes.

Commands Executed

- whoami - hostname - tasklist – systeminfo

Observed Events (SIEM)

Event 1: Timestamp: Apr 20, 2026 @ 01:12:41.721

Process: whoami.exe

Command Line: C:\Windows\System32\whoami.exe

SHA256: 1d4902a04d99e8ccbfef7085e63155955fee397449d386453f6c452ae407b8743

Event 2: Timestamp: Apr 20, 2026 @ 01:12:55.221

Process: hostname.exe

Command Line: C:\Windows\System32\hostname.exe

SHA256: N/A

Event 3: Timestamp: Apr 20, 2026 @ 01:13:04.657

Process: tasklist.exe

Command Line: C:\Windows\System32\tasklist.exe

SHA256:

be7241a74fe9a9d30e0631e41533a362b21c8f7aae3e5b6ad319cc15c024ec3f

Event 4: Timestamp: Apr 20, 2026 @ 01:13:14.080

Process: systeminfo.exe

Command Line: C:\Windows\System32\systeminfo.exe

SHA256: 6f87caa51bdea802045bb281fc2686a3c76364c26a3ffe6c2ccac4af5f9db37b

Event Analysis

Multiple enumeration commands were executed within a short time window:

- whoami.exe → identifies the current user
- hostname.exe → identifies the system
- tasklist.exe → lists running processes
- systeminfo.exe → gathers system details

This sequence strongly indicates reconnaissance activity.

Although each command can be legitimate individually, their combined execution suggests an attempt to collect system and user information.

MITRE ATT&CK Mapping Tactic:

Discovery (TA0007)

Techniques:

- T1033 – Account Discovery
- T1057 – Process Discovery
- T1082 – System Information Discovery

Detection Rules

Basic Rule:

process.name: "whoami.exe"

Improved Rule:

process.name: ("whoami.exe" OR "hostname.exe" OR "tasklist.exe" OR "systeminfo.exe")

Correlation Rule:

Multiple discovery commands executed within a short time window.

Conclusion

The activity is consistent with discovery behavior and should be considered suspicious when executed in sequence. This type of activity is commonly observed during the early stages of an attack lifecycle.

4. Lateral Movement Phase

Lateral Movement involves techniques used to move across systems and execute commands remotely.

Command Executed

WMIC.exe process call create "cmd.exe /c whoami"

Observed Events (SIEM)

Event 1: Timestamp: Apr 19, 2026 @ 19:02:58.424

Process: WMIC.exe

Command Line: "C:\Windows\System32\Wbem\WMIC.exe" process call create "cmd.exe /c whoami"

SHA256:

da603fa720ab43aa6d4d36aa9fdb828dab9645523eabaac209af6451d5b4d757

Event 2: Timestamp: Apr 19, 2026 @ 19:02:58.660

Process: cmd.exe

Command Line: cmd.exe /c whoami

SHA256: badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0

Event 3: Timestamp: Apr 19, 2026 @ 19:02:58.932Z

Process: whoami.exe

Command Line: whoami

SHA256: 1d4902a04d99e8ccbfe7085e63155955fee397449d386453f6c452ae407b8743

Event Analysis

The execution chain begins with wmic.exe using the "process call create" method to spawn a new process.

This technique allows execution through the WMI service, which can obscure the original parent process. In real-world scenarios, cmd.exe is typically spawned by WmiPrvSE.exe.

The execution of cmd.exe followed by whoami.exe confirms successful command execution via WMI.

This behavior is highly suspicious and commonly associated with lateral movement. All binaries were verified as legitimate, confirming a Living off the Land (LotL) technique.

MITRE ATT&CK Mapping Tactic:

Lateral Movement (TA0008)

Technique: T1047 – Windows Management Instrumentation

Detection Rules

Basic Rule:

process.name: "WMIC.exe"

Improved Rule: process.name: "WMIC.exe" AND process.command_line: "process call create"

Advanced Rule: process.name: "WMIC.exe" AND process.command_line: "cmd.exe"

Correlation Rule: WMIC.exe → cmd.exe → whoami.exe within a short time window.

Conclusion

This activity was classified as a True Positive and demonstrates lateral movement using WMI.

The use of legitimate system binaries confirms a Living off the Land attack pattern, which should be closely monitored in enterprise environments.

Detection Improvements

Based on the investigation of the simulated attack scenarios, detection rules were proposed and partially implemented to improve visibility and detection accuracy within the SIEM environment.

Instead of relying on isolated events, the focus was placed on identifying behavioral patterns associated with malicious activity.

Detection Rule Implementation (Elastic SIEM)

A detection rule was created in Elastic SIEM to identify suspicious access to the Windows Credential Manager using rundll32.exe.

Rule Name:

Suspicious Rundll32 Execution - Credential Access (KeyMgr)

Detection Logic (KQL): process.name: "rundll32.exe" AND process.command_line: "keymgr"

Alert Configuration:

- Time Window: 5 minutes
- Threshold: ≥ 1 event
- Grouping: host.name

Action: When triggered, the rule automatically creates a case in the SIEM, allowing analysts to track and investigate the incident.

Validation: The rule was tested by executing the simulated command, and the alert was successfully generated. Further investigation in the timeline confirmed that the process name and command line matched the expected behavior.

Summary This implementation demonstrates how identified attack behaviors can be translated into actionable detection rules.

By focusing on behavior rather than single events, detection accuracy is improved while reducing false positives.

Attack Timeline (Simulated Correlation)

[Discovery] Multiple enumeration commands observed (whoami, hostname, tasklist, systeminfo)

[Persistence] Registry Run Key modification detected via reg.exe

[Credential Access] Rundll32 used to access Windows Credential Manager

[Lateral Movement] WMIC used to execute remote command (cmd.exe /c whoami)

Conclusion: The sequence of events indicates a multi-stage attack involving post-exploitation techniques and potential lateral movement.

Analyst Decision Making From an analyst perspective:

The detection of multiple discovery commands in a short time frame would raise initial suspicion. The registry modification indicates persistence, increasing the risk level. Credential access activity suggests potential privilege escalation or data exposure. The use of WMIC to execute commands indicates possible lateral movement, escalating the incident to high severity.

Incident Response

- Isolate the affected host from the network
- Reset compromised user credentials
- Investigate potential lateral movement across other hosts
- Monitor for additional suspicious activity